

Course review

Numerical Analysis E3/I3, FMN050

Numerical Analysis

Centre for Mathematical Sciences

Lund University, Sweden

URL: <http://www.maths.lth.se/na/>

May 15, 2006

1. Error analysis

Exact computation: $y = F(x)$

Perturbed data: $y + \delta y = F(x + \delta x)$

Linearized error: $\delta y \approx F'(x) \delta x$

Error bound: $\|\delta y\| \approx \|F'(x)\| \cdot \|\delta x\|$

Note: $dy = F'(x)dx$: derivative, gradient, Jacobian

(Relative) condition number:

$$\frac{\|\delta y\|}{\|y\|} \lesssim C \cdot \frac{\|\delta x\|}{\|x\|}$$

2. Vectors, matrices, norms

Vector norm properties:

1. $\|x\| \geq 0 \quad \|x\| = 0 \Leftrightarrow x = 0$
2. $\|\alpha x\| = |\alpha| \cdot \|x\|$
3. $\|x + y\| \leq \|x\| + \|y\|$

$p = 1$	$\ x\ _1 = \sum_1^n x_i $	1-norm
$p = 2$	$\ x\ _2 = \left(\sum_1^n x_i ^2\right)^{1/2}$	Euclidean norm
$p = \infty$	$\ x\ _\infty = \max_i x_i $	max norm

Matrix norm properties:

1. $\|A\| \geq 0 \quad \|A\| = 0 \Leftrightarrow A = 0$
2. $\|\alpha A\| = |\alpha| \cdot \|A\|$
3. $\|A + B\| \leq \|A\| + \|B\|$
4. $\|AB\| \leq \|A\| \cdot \|B\|$

$p = 1$	$\ A\ _1 = \max_j \sum_i a_{ij} $	1-norm
$p = 2$	$\ A\ _2 = \sqrt{\max_i \lambda_i(A^T A)}$	Euclidean norm
$p = \infty$	$\ A\ _\infty = \max_i \sum_j a_{ij} $	max norm

3. Linear systems

LU decomposition: Factorize $A = LU$, with

- L *unit lower triangular*
- U *upper triangular*
- $n^3/3$ flops (*algorithmic complexity*)

Forward and backward substitution: Then solve

- $Ly = b$
- $Ux = y$ (implies $LUx = Ly \Rightarrow Ax = b$)
- n^2 flops (*algorithmic complexity*)

Perturbation bound for $Ax = y$:

$$\frac{\|\delta x\|}{\|x\|} \leq \|A\| \cdot \|A^{-1}\| \frac{\|\delta y\|}{\|y\|}$$

Condition number: $\kappa[A] = \|A\| \cdot \|A^{-1}\|$

4. Nonlinear equations

Fixed point iteration: $x_{n+1} = g(x_n)$

Linear convergence, only if $L[g] < 1$

Error estimate: (computable error bound)

$$\|x_{n+1} - x^*\| \leq \frac{L[g]}{1 - L[g]} \|x_n - x_{n+1}\|$$

Newton's method:

$$x_{n+1} = x_n - f(x_n)/f'(x_n)$$

Quadratic convergence, error $\varepsilon_{n+1} \sim \varepsilon_n^2$

Newton's method for systems:

$$x_{n+1} = x_n - [f'(x_n)]^{-1} f(x_n)$$

Jacobian matrix

5. Interpolation

Given data $f_i = f(x_i)$, determine a function P on the interval $[x_0, x_n]$ such that

$$P(x_i) = f_i \quad i = 0, 1, \dots, n$$

Basis functions: $\{\varphi_k(x)\}_0^n$

Determine $\{c_k\}_0^n$ from the linear system

$$c_0\varphi_0(x_i) + c_1\varphi_1(x_i) + \dots + c_n\varphi_n(x_i) = f_i$$

- Monomial basis $\varphi_k(x) = x^k$
- Newton basis
- Lagrange basis $\varphi_k(x) = \Pi'_i(x - x_i)/\Pi'_i(x_k - x_i)$
- Trigonometric interpolation $\varphi_k(x) = e^{2\pi i k x}$

6. Least squares

Determine $\{c_k\}_0^n$ from the *overdetermined* system

$$c_0\varphi_0(x_i) + c_1\varphi_1(x_i) + \cdots + c_n\varphi_n(x_i) = f_i$$

for $i = 0, \dots, m > n$

$$Ac \approx f$$

Normal equations: $A^T Ac = A^T f$

- The solution minimizes the residual $\|Ac - f\|_2$
- The residual is orthogonal to the columns of A
- Basis functions could have any preferred form

7. Numerical integration

Approximate f by interpolation polynomial

$$\int_a^b f(x) \, dx \approx \int_a^b P(x) \, dx = \sum_{i=0}^n w_i f(x_i)$$

Trapezoidal sum $T(\Delta x)$:

$$\begin{aligned} \int_{x_0}^{x_N} f(x) \, dx &\approx \sum_{i=0}^{N-1} \frac{1}{2} (f_i + f_{i+1}) \Delta x = T(\Delta x) \\ &= \Delta x \cdot \left(\frac{f_0}{2} + f_1 + \cdots + f_{N-1} + \frac{f_N}{2} \right) \end{aligned}$$

Simpson sum $S(\Delta x)$:

$$S(\Delta x) = \frac{\Delta x}{3} (f_0 + 4f_1 + 2f_2 + 4f_3 + \cdots + 4f_{2m-1} + f_{2m})$$

Numerical integration. . .

Method order: If exact for $\deg P = p - 1$, then the method's order is p .

Trapezoidal rule: $p = 2$

Simpson sum: $p = 4$

Trapezoidal rule:

$$T(\Delta x) = T(0) + c_1 \Delta x^2 + c_2 \Delta x^4 + c_3 \Delta x^6 + \dots$$

Romberg's method:

Trapezoidal rule with repeated Richardson extrapolation

8. Richardson extrapolation

For successive elimination of error terms use corrections

$$\frac{\Delta}{q^{p_i} - 1}$$

Example 1: Trapezoidal rule

$T(h) = y(0) + c_1h^2 + c_2h^4 + c_3h^6 + \dots$ with step doubling, $q = 2$, gives correction terms

$$\Delta/3, \quad \Delta/15, \quad \Delta/63, \quad \dots$$

Example 2: 2p-BVP with symmetric finite differences. Order $p = 2$ without correction, $p = 4$ after one correction, $p = 6$ after two corrections, etc.

Example 3: Eigenvalues in Sturm-Liouville problem

$$\lambda(h) = \lambda + c_1h^2 + c_2h^4 + c_3h^6 + \dots$$

9. Eigenvalue computation

$$Ax = \lambda x; \quad x^H x = 1$$

Power iteration algorithm:

- Starting value y_0
- Iterate until convergence:

$$\hat{y}_k := y_k / \|y_k\|_2$$

$$y_{k+1} := A\hat{y}_k$$

$$\sigma_k := \hat{y}_k^H y_{k+1}$$

Inverse iteration algorithm:

- Starting value y_0
- Choose shift s
- LU -factorize $A - sI$
- Iterate until convergence:

$$\hat{y}_k := y_k / \|y_k\|_2$$

$$(A - sI)y_{k+1} := \hat{y}_k$$

$$\sigma_k := \hat{y}_k^H y_{k+1}$$

10. Initial value problems

Standard formulation

$$\dot{y} = f(t, y); \quad y(0) = y_0$$

Explicit Euler method:

$$y_{n+1} = y_n + hf(t_n, y_n)$$

Implicit Euler method:

$$y_{n+1} = y_n + hf(t_{n+1}, y_{n+1})$$

Requires equation solving every step

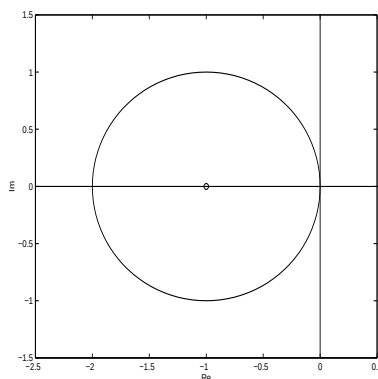
Order $p = 1$ (exact for polynomials of degree 1)

Stability analysis

“Test equation”: $\dot{y} = \lambda y$

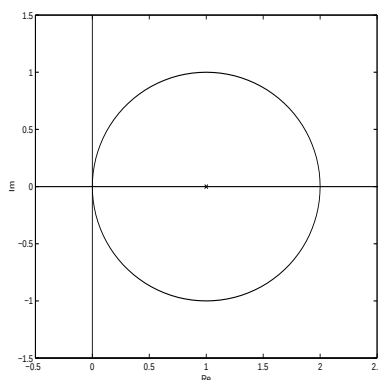
Explicit Euler: $y_{n+1} = (1 + h\lambda)y_n$

Stability region: $\mathcal{S} = \{h\lambda \in \mathbb{C} : |1 + h\lambda| \leq 1\}$



Implicit Euler: $y_{n+1} = (1 - h\lambda)^{-1}y_n$

Stability region: $\mathcal{S} = \{h\lambda \in \mathbb{C} : |1 - h\lambda| \geq 1\}$



Stiffness

Solve $\dot{y} = \lambda(y - \sin t) + \cos t$ **with** $\lambda \ll -1$.

Solutions:

Particular: $y_P(t) = \sin t$

Homogeneous: $y_H(t) = e^{\lambda t}$

General: $y(t) = e^{\lambda(t-t_0)}(y(t_0) - \sin t_0) + \sin t$

Homogeneous solutions decay very fast when $\lambda \ll -1$

Only implicit methods are stable for $h\lambda \ll -1$

Explicit methods must use steps with $|h\lambda| \ll 1$

Stiffness: explicit methods are limited to very short steps to remain stable

Runge–Kutta and multistep methods

RK4 Order $p = 4$.

$$\dot{Y}_1 = f(t_n, y_n)$$

$$\dot{Y}_2 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}\dot{Y}_1\right)$$

$$\dot{Y}_3 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}\dot{Y}_2\right)$$

$$\dot{Y}_4 = f(t_n + h, y_n + h\dot{Y}_3)$$

$$y_{n+1} := y_n + \frac{h}{6} (\dot{Y}_1 + 2\dot{Y}_2 + 2\dot{Y}_3 + \dot{Y}_4)$$

Multistep method (BDF2) Order $p = 2$

$$\frac{3}{2}y_{n+1} - 2y_n + \frac{1}{2}y_{n-1} = hf(t_{n+1}, y_{n+1})$$

11. Two-point boundary value problems

$$y'' = f(x, y)$$

$$y(0) = \alpha; \quad y(1) = \beta$$

Introduce equidistant grid with $\Delta x = 1/(N + 1)$.

Discretize:

$$\frac{y_{i+1} - 2y_i + y_{i-1}}{\Delta x^2} = f(x_i, y_i)$$

$$y_0 = \alpha; \quad y_{N+1} = \beta$$

N nonlinear equations for N unknowns $y_1, y_2, \dots, y_N$.

Solve using Newton's method.

Sturm–Liouville problem

Discretization of $y'' + \lambda y = 0$ with boundary values $y(0) = y(1) = 0$

$$\frac{y_{i-1} - 2y_i + y_{i+1}}{h^2} + \lambda_h y_i = 0$$

$$y_0 = y_{N+1} = 0; \quad h = 1/(N + 1).$$

Tridiagonal $N \times N$ matrix formulation

$$\frac{1}{h^2} \begin{pmatrix} -2 & 1 & & \\ 1 & -2 & 1 & \\ & & \ddots & \\ & & 1 & -2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{pmatrix} = -\lambda_h \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{pmatrix}$$

Algebraic eigenvalue problem

$$T_h y = -\lambda_h y$$

12. Discrete Fourier Transform

Sequence of function values: $\mathbf{f} = \{f(n/N)\}_{n=0}^{N-1}$

Fourier coefficient vector: $\mathbf{c} = \{c_m\}_{m=0}^{N-1}$

Introduce the $N \times N$ **matrix**

$$\mathcal{F}_N = \left\{ \frac{e^{-2\pi i m n / N}}{N} \right\}_{m,n=0}^{N-1}$$

Then the matrix–vector multiplication

$$\mathbf{c} = \mathcal{F}_N \cdot \mathbf{f}$$

is the **Discrete Fourier Transform (DFT)**

Fast Fourier transform, FFT:

If $N = 2^p$ then $N \log_2 N$ operations only